Date:

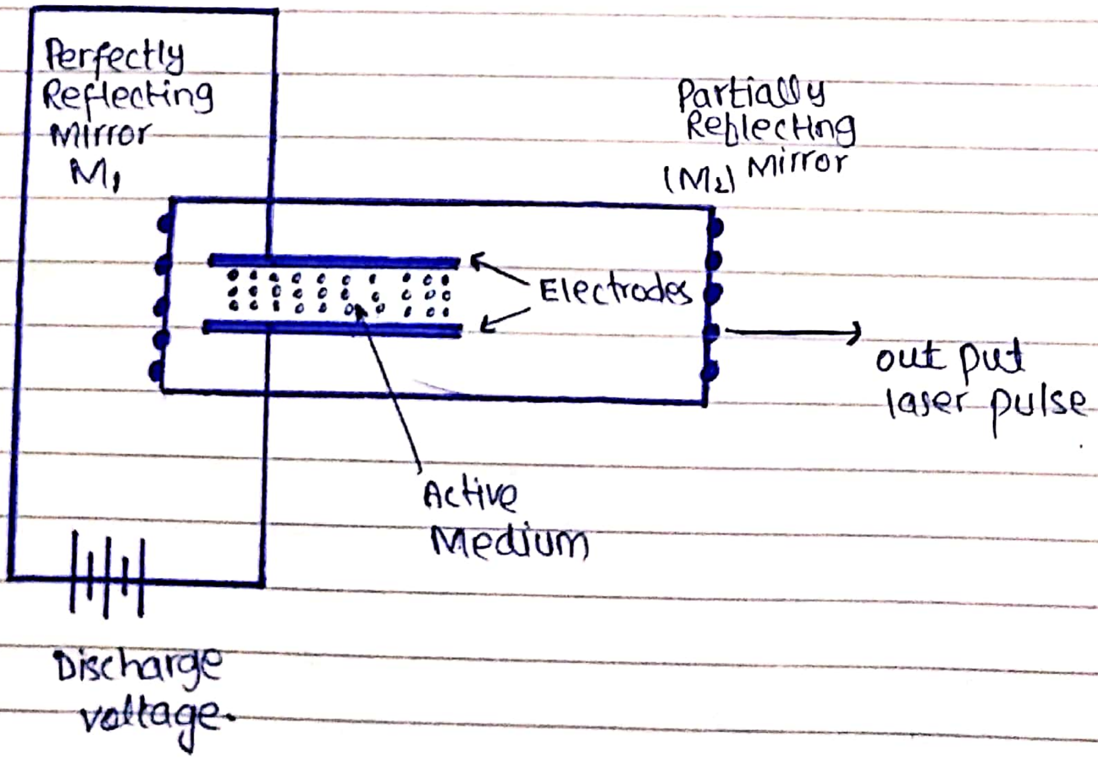
Nitrogen Laser:

In nitrogen laser, active or gain or lasing medium in nitrogen molecules in the gaseous phase. It is three level laser.

Population inversion in nitrogen laser is acheived by electrical Pumping source. Normally Pumping is provided by direct electron impact.

Construction:

The arrangement to produce a nitrogen pulsed laser is shown in the figure.



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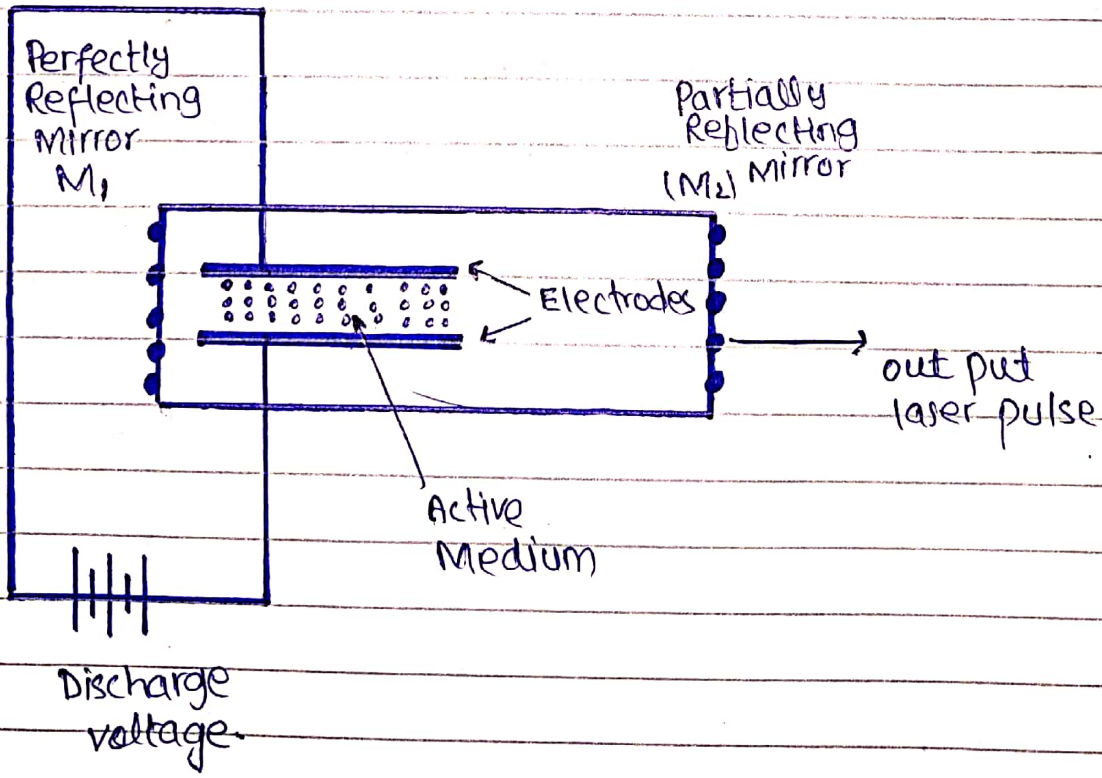

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It consist of a tube formed by two electrodes connected to the discharge voltage source (10–40 KeV).

The active medium in the form of Nitrogen molecules in the gaseous phase is placed between these two electrodes.

The arrangement is enclosed in a resonant cavity formed by two planes and parallel mirrors M_1 and M_2.
Mirror M_1 is perfectly reflecting mirror while mirror M_2 is partially reflecting mirror through which pulse the laser comes out.

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Working:—

Active Medium (or Nitrogen Laser gain Medium)

The main component of a Nitrogen laser is a medium in the form of a Nitrogen molecules called active medium. The main characterstics of the active medium are as follows:

* It must have a Pair of energy levels separated by certain amount of energy. The energy level having energy is Known as upper energy level and energy level having low energy is Known as low energy or ground state.

* It must allow a Population inversion between two energy levels.

Pumping Device:—

It is an external source of energy That Provide the necessary energy to the active medium to Produce a state of Population inversion essential for laseing action.

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A Resonant Cavity:

Population inversion is acheived to amplified the signal by (or Proton) Stimulated emission. Most of the atoms in the excited state emit spontanously and do not contribute to overall out put.

Only a few atoms in the excited state emit via stimulated emission and hands overall gain of the out put is small.

A resonant cavity consist of a pair of a plane or spherical Mirrors placed parallel to each other at the end of active medium. One of the mirror is fully reflecting mirror and other is partially transmitting mirror. The laser out put is taken out through the partially transmitted mirror which is also called **output coupler mirror.**

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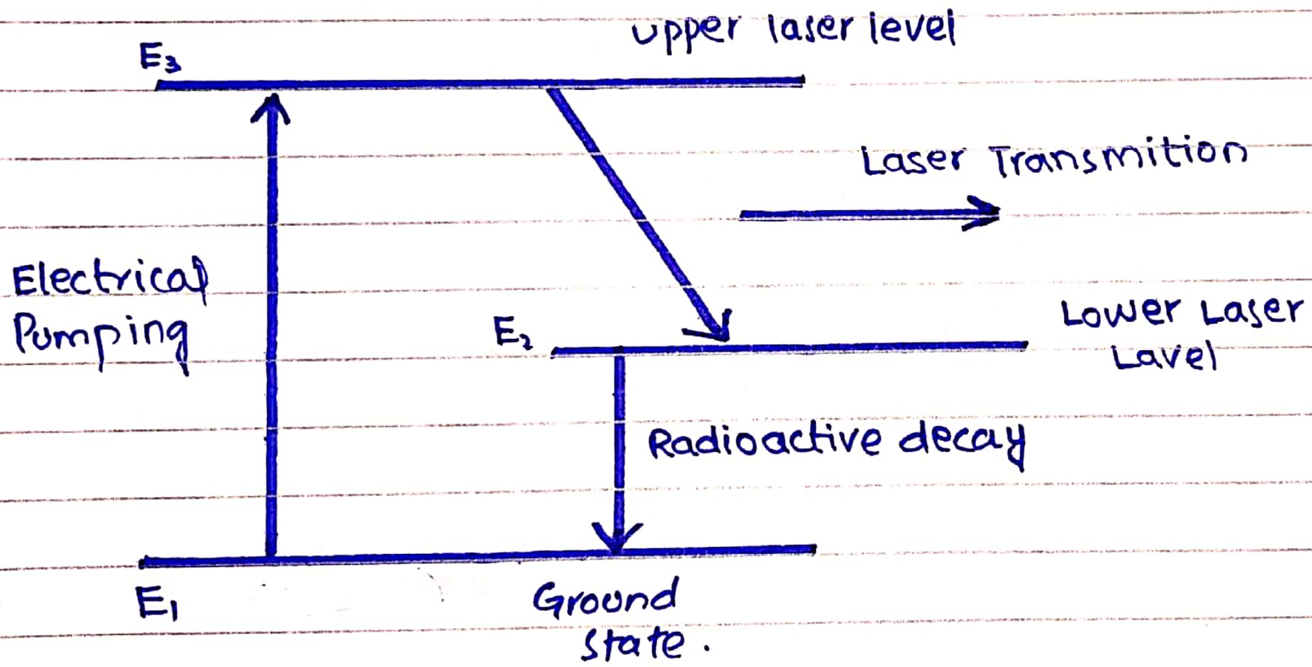
Energy Level Diagram:-

The energy level diagram of the Nitrogen molecules in the gaseous Phase is shown in the figure.

Each energy level consist of series of vibrational energy levels depending on the internuclear separations.



Nitrogen molecule is excited from the ground state E_1 to the excited state energy level$_3$ (called upper laser level ULL) by direct collision with electrons in the discharge tube. Nitrogen molecules falls from Energy level E_3 level E_2 (called lower laser level LLL) by emitting a Photon of ultravoilet light of wavelength 337.1 nm.

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The lifetime of upper energy level is less Than lifetime of lower energy level. The output of Nitrogen laser is in The form of Pulses i.e Nitrogen laser is a Pulsed laser.

In Nitrogen laser the width of laser Pulse is narrow. It is because when a laser transmition take Place, The Population of upper laser level E_2 decreases, and the Population of lower laser lavel E_1 increases. The Nitrogen molecule than falls from energy level E_2 to the ground state and laser action stops. Thats why Nitrogen laser is known as Self Terminating laser.

Disadvantages :-

* The beam quality of the Nitrogen laser is Poor.
* The discovery of Nitrogen laser is large as compared to divergence of the laser.
* The out put Power is low.
* The efficiency of Nitrogen laser is low.

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Applications :-

Nitrogen laser are used in :

1) Pumping of Dye laser

2) Measuring Air Pollution

3) Scientific Research

4) Spectroscopy and fluorescence studies

5) Fast Speed Photography.